Solution Architecture

Date	30 June 2026
Team ID	xxxxxx
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer
[e.g., Web App (React), Mobile App (Flutter), Admin Dashboard]
User Interface for Weather Data Input Prediction Results Dashboard

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API Gateway / Load Balancer
[e.g., Nginx, AWS API Gateway, Express Server] Receives user requests, validates input, and forwards data to the prediction engine.

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Auth Service
[e.g., JWT, OAuth2]

Core Logic Service
[e.g., Data Processing, Machine Learning Models]

External API Integrations
[e.g., Payment Gateway, Email Service]

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Data / Storage Layer
[e.g., PostgreSQL Database, MongoDB, AWS S3]

Component Description Table

Component Name	**Description / Role in Architecture**	**Technologies Used**
Presentation Layer	*Provides a user-friendly web interface where users enter weather parameters and view flood prediction results.*	*Flask, HTML, CSS, JavaScript*
API Gateway / REST API	*Receives user requests, validates inputs, and forwards the data to the machine learning prediction engine. Returns prediction results to the web application.*	*Flask REST API, Python*
Core Logic Service	*Performs data preprocessing, feature engineering, model loading, and flood prediction using trained machine learning algorithms.*	*Python, Scikit-learn, XGBoost, Pandas, NumPy*
External API Integrations	*Retrieves weather information from external sources (optional) and supports cloud deployment services for future scalability.*	*IMD Weather API (Optional), IBM Cloud*
Data / Storage Layer	*Stores the dataset, trained model, and prediction logs for the application.*	*Kaggle Dataset, Joblib (.pkl), SQLite / CSV*